Deep Q-Network

CartPole-v1 and ALE/MsPacman-v5

GitHub SAMSUNYUJUN / mlds-rl-assignment2
https://github.com/SAMSUNYUJUN/mlds-rl-assignment2

Environment	Training episodes	Selected checkpoint	500-episode mean ± SD
CartPole-v1	500	180	500.00 ± 0.00
ALE/MsPacman-v5	4,000	2,250	2,658.32 ± 601.03

Both agents used an online action-value network, a uniformly sampled replay buffer, and a lagged target network. The experiments used a single NVIDIA RTX A6000 on DeepDish.

Learning rule

The network combines a state-value head with an action-advantage head after subtracting the mean advantage. At each update the online network selects the next action and the target network evaluates it. Adam minimizes the Huber loss between Q(s, a) and the following target.

$$y = \Sigma_{j=0}^{k-1} \gamma^j r_j + d\, Q_{target}(s', \operatorname{argmax}_{a'} Q_{online}(s', a'))$$

The multiplier d is γ^k while continuation is possible and 0 at a learning terminal. Each lane has its own queue of up to three transitions. The learner drains shorter tails at boundaries, so the target never includes a reward from the following episode or life.

Exploration starts with uniform actions. Updates begin after a replay warmup, and ε then decreases linearly with agent steps. Replay samples only filled slots and overwrites the oldest entries when full. The learner excludes Gymnasium's reset-only steps from replay, Q statistics, and episode returns.

Evaluation protocol

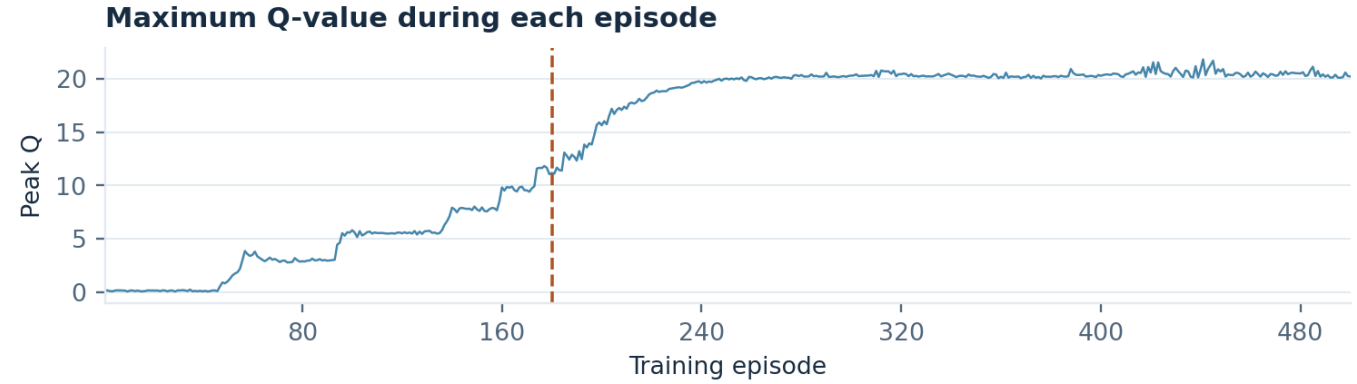
Periodic greedy rollouts on fixed selection seeds choose the checkpoint with the highest mean raw reward. Ties keep the earlier checkpoint. Each final evaluation uses ε = 0 and 25 newly seeded environment streams with 20 episodes per stream. The tables and histograms use raw rewards and population SD. Policy probes use a third seed range.

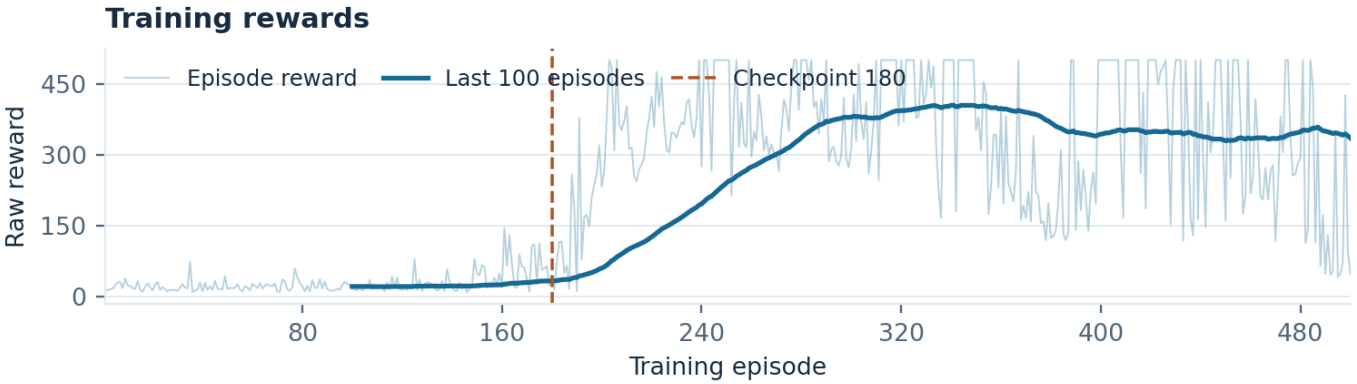
The Q plots show the largest estimated Q over all actions and visited states before actions in each training episode. The 100-episode moving averages begin at episode 100. The orange marker locates the selected checkpoint.

CartPole-v1

Training and 500 greedy evaluation episodes

The agent observes cart position, cart velocity, pole angle and pole angular velocity and chooses left or right. Input scales are 2.4, 2, π/15 and 2 in the environment's observation order. A 128-unit encoder feeds two 128-unit heads. The discount is 0.95.







Evaluation. Mean 500.00, SD 0.00, range 500 to 500. The pole remained upright through the 500-step limit in 500/500 episodes. Checkpoint 180 was selected using 80 evaluation seeds.

The first and final 100 training episodes averaged 21.36 and 335.23. The final frozen policy averaged 27.29 in the paired probe, which shows that continued updates can degrade a policy. At $\gamma = 0.95$, even 150 steps already yield nearly 20 discounted reward, so high Q alone gives little evidence of 500-step balance.

CartPole validation

First-action interventions and the 500-step boundary

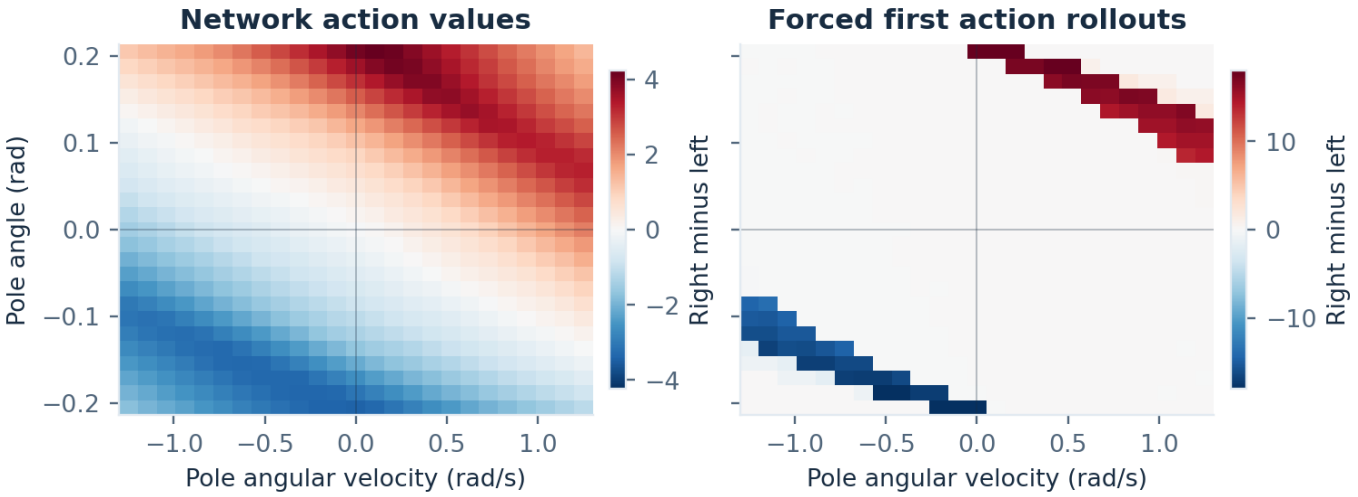


Figure. For each of 625 fixed pole states with the cart centered and still, the experiment forces left or right once, then runs the same greedy controller. Red favors right, blue favors left. The right panel uses realized discounted return through termination or 500 steps.

Among the 95 states with an absolute return gap of at least 0.10, the network preferred the better first action in 95. Following the predicted first action lost 0.000024 discounted reward on average across all states, with worst loss 0.0031. This grid includes angles and velocities beyond the normal reset distribution.

Policy	Mean steps	Selected minus policy (95% paired CI)
Uniform random	22.25	+477.75 [+474.00, +481.15]
Untrained greedy	9.50	+490.50 [+490.27, +490.73]
Final checkpoint (500)	27.29	+472.71 [+471.33, +474.04]
Selected greedy	500.00	Reference policy

The comparison uses 48 common reset seeds. The intervals resample paired seed differences 10,000 times.

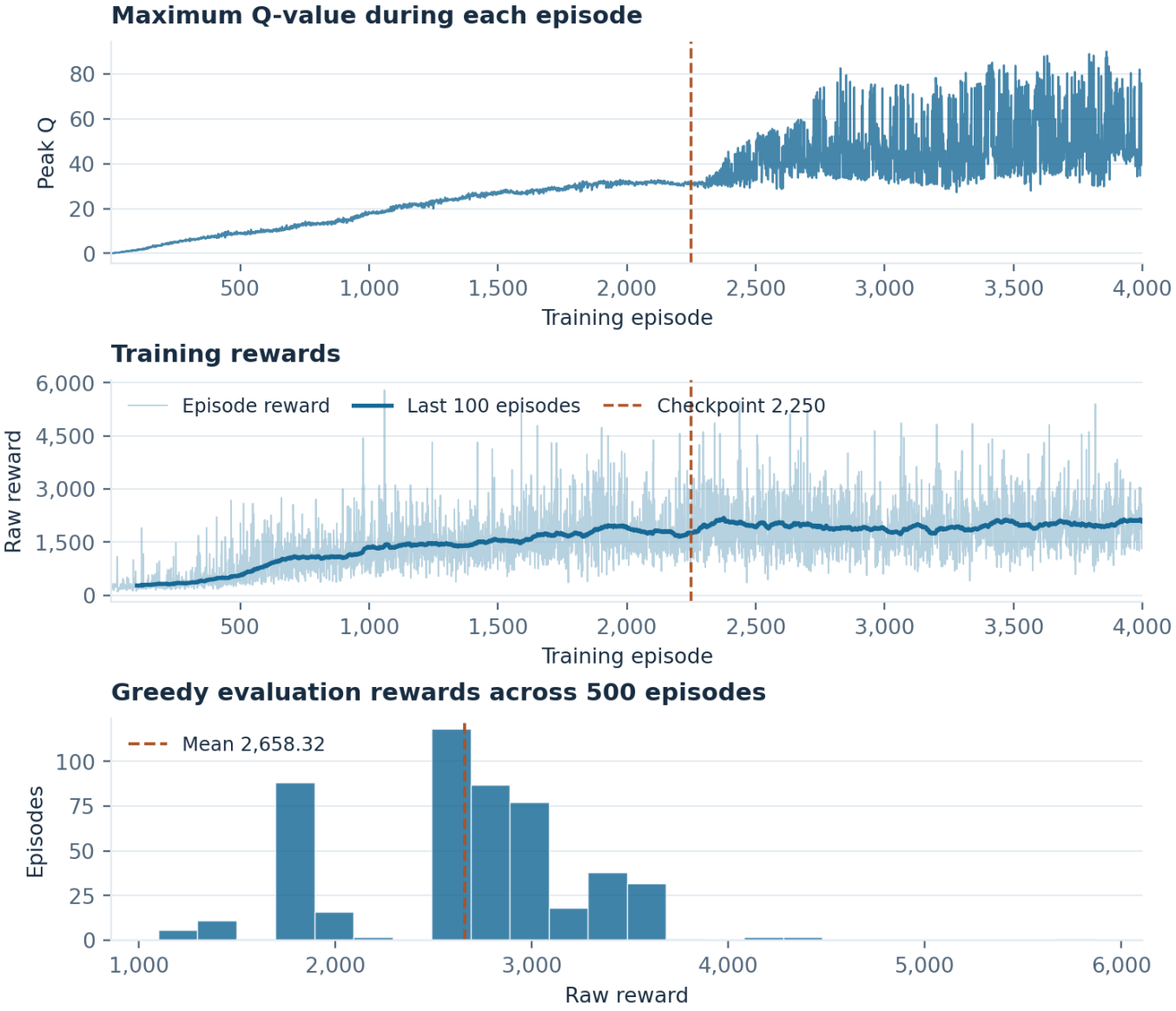
Physical failure and the time cap

A seeded three-step rollout and an otherwise identical capped rollout reached the same final state. The cap kept the continuation term, giving a one-step target of 11.346. Treating the cap as physical failure would give 1.0. A forced angle failure from $\theta = 0.208$ rad and angular velocity 0.4 rad/s produced target 1.0 with zero continuation. Physical failure takes precedence when both end flags appear at step 500.

MsPacman-v5

Training and 500 greedy evaluation episodes

ALE supplies four stacked 84 × 84 grayscale observations with max pooling. The CNN uses three convolutions and two 128-unit heads to score nine actions. The v5 emulator retains four-frame action repeat and 25% sticky actions. The discount is 0.99 and replay uses raw points divided by 10.



Evaluation. Mean 2,658.32, SD 601.03, median 2,700 and range 1,100 to 5,870. The selected checkpoint followed episode 2,250.

The first and final 100 training episodes averaged 276.40 and 2,085.00 raw points. The peak Q uses scaled, discounted rewards and per-life bootstrap boundaries. The raw score curves and fixed-seed greedy comparisons measure policy improvement directly.

MsPacman validation

Motion information, score progress and life boundaries

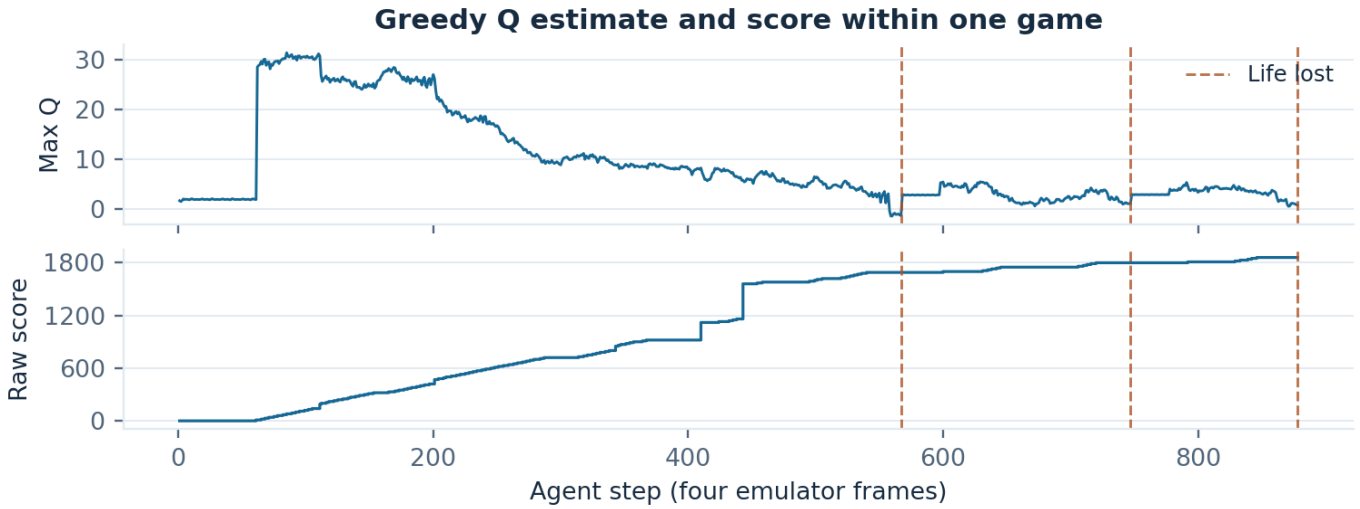


Figure. The selected policy on seed 300,100. Orange marks life losses. The Q trace gives the pre-action estimate and the score trace includes the reward earned on that action.

Policy	Mean points	Selected minus policy (95% paired CI)
Uniform random	245.42	+2,340.00 [+2,173.33, +2,503.14]
Untrained greedy	70.00	+2,515.42 [+2,353.12, +2,676.67]
Early checkpoint (250)	358.96	+2,226.46 [+2,084.17, +2,375.21]
Final checkpoint (4,000)	2,247.08	+338.33 [+178.33, +499.39]
Selected with repeated newest frame	1,532.71	+1,052.71 [+841.67, +1,261.46]
Selected with frame history	2,585.42	Reference policy

The 48 seed pairs use identical reset seeds and v5 settings. Repeating the newest frame removes the temporal differences the trained CNN receives. The intervention also changes the input distribution. Retraining with a single-frame input would isolate the benefit of frame history.

Life loss, game over and the frame cap

The replay tail drains with zero bootstrap at a lost life at action 567. The game continues with the same score accumulator. In the recorded trajectory, the replay returns match an independent backward calculation at all 3 life and episode boundaries. On the same trajectory capped at 244 emulator frames, the replay tail retained returns (0.9801, 0.99, 1) and bootstrap multipliers (0.970299, 0.9801, 0.99). Reset-only steps are excluded from the traces and replay.

Reproduction

Commands, parameter settings and further validation

Run the following commands from a DeepDish login. Set A2_GPU to the assigned GPU index. The script trains both policies, evaluates 500 episodes per policy, and saves all validation data. Use a fresh runs directory for each full run.

```
git clone https://github.com/SAMSUNYUJUN/mlds-rl-assignment2.git
cd mlds-rl-assignment2
python3 -m venv --system-site-packages .venv
.venv/bin/python -m pip install -r requirements.txt
A2_GPU=0 bash deepdish.sh
.venv/bin/python make_report.py --cartpole runs/cartpole --mspacman runs/mspacman
```

Setting	CartPole-v1	ALE/MsPacman-v5
Training seed	137	42
Parallel environments	8	24
Replay capacity / warmup	50,000 / 1,024	75,000 / 12,000
Batch size / update interval	64 / 4	64 / 12
Adam learning rate	0.0005	0.00015
Target copy (updates)	250	800
ε decay (steps) / floor	12,000 / 0.03	550,000 / 0.05
N-step horizon	3	3

CartPole ran 115,429 agent steps with PyTorch 2.6.0+cu124 on NVIDIA RTX A6000. MsPacman ran 3,045,201 agent steps with PyTorch 2.6.0+cu124 on NVIDIA RTX A6000. README.md documents individual commands and the recorded files.

Experiments with three more months

Month 1. Repeat training over at least ten seeds, compare one-step and three-step targets at matched environment steps, and report between-seed variation in time to solve CartPole. Sweep cart position and velocity as well as pole state, then test sustained balance beyond the nominal cap.

Month 2. Retrain MsPacman with one versus four frames, with and without life-loss bootstrapping, under equal frame and update budgets. Track score per life, maze clears, time to first death and the median score alongside the mean to measure the effect of rare bonus rewards.

Month 3. Use fresh seed sets and vary stickiness and initial no-op counts. For both games, compare predicted Q to empirical discounted returns from cloned states under each available first action. Report value bias, action regret and bootstrap intervals over independent training seeds.

Mnih et al. (2015), DQN · van Hasselt et al. (2016), Double DQN · Wang et al. (2016), dueling networks
CartPole environment · ALE MsPacman environment